

EVENTEASE: AI-POWERED COLLEGE EVENT MANAGEMENT SYSTEM

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DECLARATION

We hereby declare that the project report entitled “**EventEase: AI-Powered college Event Management System**”, submitted to the **Institute of Engineering & Technology, Lucknow, Computer Science Engineering Department**, is a genuine record of the original work carried out by us during the course of our studies. The work presented in this report is the result of our own effort, research, and analysis completed under the valuable guidance and supervision of **Prof. Vineet Kansal and Mr. Abhishek Singh**.

We further affirm that this work has **not been submitted**, in part or whole, to any other university, institute, or organization for the award of any degree, diploma, certificate, or any other academic recognition. All information, data, diagrams, software modules, and materials used in this report are based on our own contributions, except where specific references or acknowledgments have been provided.

We also confirm that we have adhered to the ethical norms of academic integrity. Any sources, tools, or external materials consulted have been appropriately cited to the best of our knowledge. We fully understand that any instance of plagiarism, fabrication of data, or violation of academic honesty may lead to disciplinary action as per the rules and regulations of the institution.

We take full responsibility for the authenticity and accuracy of the content presented in this report.

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CERTIFICATE

This is to certify that the Project Report entitled “**EventEase: AI-Powered college Event Management System**”, submitted by **Kanhaiya Mittal (2200521520026)**, **Pulkit Diwakar (2300520109005)** and **Vikas Chaudhary (2300520109006)**, in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology (B.Tech.)** in the **Department of Computer Science and Information Technology, Institute of Engineering & Technology, Lucknow**, is a bona fide record of the work carried out by the candidates under my supervision and guidance.

The work presented in this report is the result of their sincere effort, technical understanding, and active participation throughout the project duration. To the best of my knowledge, the contents of this report are original, and the work embodied herein has not been submitted, either in part or full, to any other institute or university for the award of any degree, diploma, or any other academic recognition.

The candidates have shown dedication, discipline, and a clear understanding of the concepts involved in the successful completion of this project. I commend their effort and wish them success in all their future academic and professional endeavors.

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ABSTRACT

EventEase: AI-Powered College Event Management System is a modern web-based platform developed to address the increasing complexities of managing events within educational institutions. Colleges frequently conduct technical fests, cultural events, seminars, workshops, hackathons, and extracurricular programs. However, the existing manual or semi-digital methods still depend heavily on paperwork, offline registrations, printed certificates, and manual coordination. These lead to inefficiencies, data inaccuracies, delays, and mismatches between student expectations and organizational capabilities. EventEase aims to overcome these challenges using advanced technologies such as the MERN stack—MongoDB, Express.js, React.js, and Node.js—combined with artificial intelligence (AI) modules for recommendation and automated certificate generation.

The system centralizes the entire lifecycle of event management, beginning with event creation by organizers and approval by administrators. Students can view upcoming events, register digitally, track event status, and receive automated notifications. Organizers can track registrations in real-time, manage event updates, communicate with participants, and simplify attendance tracking using QR-code scanning. After the event, AI automatically generates personalized certificates with accurate details, significantly reducing the manual workload for coordinators. Administrators gain a comprehensive dashboard to monitor all activities, user roles, events, analytics, and platform security.

EventEase incorporates a clean UI, role-based authentication, secure database operations, cloud storage, and responsive design, ensuring accessibility across devices. It enhances communication, eliminates redundant work, and introduces transparency across all event-related operations. The use of AI enables intelligent sorting of events, smart suggestions for students based on interests or past participation, and reducing manual errors in certificate generation. The system also maintains a secure, structured, and scalable database capable of handling large volumes of registrations, attendance logs, and certificate records.

By providing automation, digitization, and AI-driven workflows, EventEase reduces human dependency, improves organizational efficiency, and ensures timely event execution. Its scalable architecture allows future enhancements such as mobile app development, payment gateway integration for paid events, and predictive analytics for event turnout. Ultimately, EventEase transforms traditional event management into a seamless, paperless, and systematic digital experience, benefiting students, organizers, and administrators alike.

INTRODUCTION

1. Purpose

The primary purpose of EventEase is to **transform traditional college event management into a fully digital and automated system** that enhances efficiency, accuracy, and user experience. The project aims to reduce the dependence on manual processes and introduce intelligent workflows that simplify the responsibilities of students, organizers, and administrators.

One of the key purposes is to **centralize all event-related information** in one platform so that students can easily discover upcoming events, read details, register instantly, and stay updated throughout the event lifecycle. This reduces communication barriers and ensures equal access to information for all students.

The platform also aims to **improve transparency and monitoring** by providing administrators with real-time dashboards. Through these dashboards, administrators can approve events, track registrations, monitor attendance, view analytics, and ensure smooth functioning of the overall system.

2. Background

Event management is an essential part of academic institutions, contributing to the holistic development of students through workshops, seminars, cultural festivals, technical fests, competitions, and other co-curricular activities. However, despite the growing scale and frequency of such events in colleges, the management process continues to rely heavily on outdated manual practices. Traditionally, colleges handle event-related tasks through handwritten registration forms, printed notices, Excel spreadsheets, manual attendance sheets, and individually prepared certificates. These methods are not only time-consuming but also prone to human errors, miscommunication, and data loss.

Another major challenge in the existing system is **lack of integration**. Event announcements are made on notice boards or through classroom announcements, which often fail to reach all students. Registrations collected through forms or Google Sheets become difficult to manage when the number of participants increases. Attendance tracking using manual signatures introduces errors and makes it hard to verify authenticity. Additionally, certificate preparation after events requires significant time and effort, especially for large-scale events involving hundreds of participants.

With the rise of digital transformation in educational institutions, there is an increasing need to shift from traditional manual methods to automated, intelligent, and efficient systems. Modern technologies such as cloud computing, AI automation, QR-code systems, and full-stack web development frameworks provide an opportunity to build a platform that can simplify these processes.

3. Objectives

- **Role-Based Access Control (RBAC) for Secure System Usage:** The system aims to provide role-based access to different users such as **students, event organizers, and administrators**, ensuring that each user can access only the features relevant to their role. This enhances system security, prevents unauthorized access, and ensures smooth management of events and data.
- **Digitalization of Event Creation, Registration, and Approval:** The platform will completely digitalize the event management workflow by enabling organizers to create and manage events online, students to register seamlessly, and administrators to approve or reject events through a centralized interface. This eliminates manual paperwork and significantly reduces processing time.
- **I-Based Certificate Generation and Distribution:** The system will implement **AI-powered certificate generation**, allowing certificates to be automatically created after successful event participation. This ensures accuracy in participant details, reduces human error, and enables quick digital distribution of certificates to students.
- **QR-Code-Based Attendance Management:** To minimize manual attendance marking, the system will use **QR-code-based attendance scanning**. Each participant will scan a unique QR code during the event, ensuring faster attendance recording, reduced fraud, and improved data accuracy.
- **Student-Friendly Event Discovery and Certificate Access:** Students will be able to easily explore upcoming and ongoing events through an intuitive interface, receive real-time notifications regarding registrations, approvals, and certificates, and access all their earned certificates in one secure digital repository.
- **Administrative Monitoring Through Dashboards and Analytics:** Administrators will have access to **comprehensive dashboards and analytics**, providing real-time insights into events, registrations, attendance, and certificate distribution. This helps in effective monitoring, decision-making, and overall system governance.
- **Paperless, Efficient, and Scalable System Using Modern Technologies:** The project aims to develop a **fully paperless, efficient, and scalable platform** using modern technologies such as the **MERN stack (MongoDB, Express.js, React.js, Node.js)** along with AI components. The system will be designed to handle a large number of users and events while maintaining performance and reliability.

MOTIVATION

The motivation behind EventEase is to transform the conventional event management system into a **smart, digital, and automated platform** that simplifies the entire event lifecycle. By centralizing event creation, registration, approval, attendance tracking, and certificate distribution into a single platform, EventEase aims to eliminate redundancy, reduce manual effort, and improve accuracy across all processes.

Another key driving factor is the need to enhance **user experience** for all stakeholders. Students require a simple and accessible platform to explore events, register instantly, receive notifications, and securely access certificates. Organizers need efficient tools to manage events without administrative complexity, while administrators require real-time monitoring, analytics, and control to ensure smooth and transparent operations. EventEase addresses these needs through role-based access, intuitive interfaces, and real-time dashboards.

Additionally, the project is motivated by the potential of modern technologies such as **full-stack web development (MERN stack), AI automation, and QR-code systems**. These technologies enable the creation of a scalable, paperless, and intelligent system capable of handling large numbers of users and events with reliability and speed. Integrating AI for automated certificate generation and QR-based attendance further enhances accuracy and reduces operational delays.

Overall, EventEase is motivated by the vision of creating a **future-ready event management system** that not only improves administrative efficiency but also enhances student engagement, ensures transparency, and supports sustainable, paperless practices within academic institutions.

PROBLEM STATEMENT

1. Introduction to the Problem

Event management plays an essential role in colleges, involving activities such as seminars, fests, workshops, and cultural programs. However, many institutions still rely on outdated manual methods for handling these events. This creates inefficiencies, delays, and a lack of coordination among students, organizers, and administrators.

2. Issues in the Current Event Management Process

Event announcements are commonly made through notice boards, class visits, or informal messaging. This makes it difficult to reach all students on time. Registrations are collected using paper forms, Google Forms, or Excel sheets, which become difficult to manage, especially during large events.

3. Limitations of Manual Methods

Manual attendance through signature sheets is slow, inaccurate, and prone to proxy entries. Certificate creation requires manually typing names, verifying details, and printing hundreds of certificates—leading to errors, delays, and unnecessary workload. Storing and managing event-related data manually also risks data loss or inconsistency.

4. Challenges Faced by Students and Organizers

Students often miss event information due to poor communication. Organizers struggle with handling large participant data, tracking attendance, and generating certificates on time. Administrators face difficulty monitoring events across different departments because there is no unified platform to view all activities.

5. Need for an Automated Solution

Due to the limitations of manual processes, there is a clear need for a centralized, digital, and automated event management system. A platform like EventEase can streamline event creation, registration, attendance, and certificate distribution while reducing errors, improving communication, and increasing efficiency for all stakeholders.

6. Data Management and Record-Keeping Issues

Colleges struggle to maintain accurate records of registrations, attendance, and certificates. Manual data storage makes it difficult to retrieve past records, generate reports, or verify participation for audits or student portfolios.

7. Lack of Real-Time Monitoring and Analytics

Administrators and organizers have no tools to track event performance in real time. They cannot view how many students registered, who attended, or which events are gaining popularity. This lack of analytics leads to poor decision-making and planning.

8. Communication Gaps and Delays

Students often miss updates about schedule changes, event reminders, or certificate availability due to inefficient communication methods. Organizers also face delays in reaching participants, leading to confusion and lower participation.

LITERATURE REVIEW

A literature review presents an overview of existing research, systems, frameworks, and technologies related to event management automation and AI-driven academic platforms. The development of EventEase involved analyzing traditional event management practices, currently available digital tools, their limitations, and emerging technologies in web development and artificial intelligence [1], [2].

Traditional event management in educational institutions primarily relies on manual registration processes, physical attendance tracking, offline announcements, and printed certificate distribution. Studies indicate that such approaches are error-prone, time-consuming, and lack consistency across departments. Researchers emphasize that digitization in academic administration significantly reduces administrative workload, enhances student participation, and improves operational efficiency [1], [4].

Several digital tools such as Google Forms, Microsoft Excel, and standalone event management applications have been introduced to simplify event registration. However, literature highlights that these tools offer only partial automation and lack real-time analytics, role-based access control, and seamless institutional integration. Furthermore, most systems do not support the complete event lifecycle, particularly automated attendance tracking and certificate generation [2], [6].

Recent academic research discusses the integration of artificial intelligence in administrative and academic systems. AI-based platforms can analyze historical participation data, recommend relevant events to students, and automate document generation. Studies on AI-powered certificate generation report substantial improvements in accuracy and processing time, especially for large-scale events. Additionally, student feedback in multiple studies indicates a strong preference for automated systems that provide instant certificate access without administrative delays [3], [5].

The MERN (MongoDB, Express.js, React, Node.js) stack is widely acknowledged in the literature as an effective framework for developing scalable and efficient web applications. Research highlights MongoDB's flexibility in managing event-centric data, React's capability to build dynamic user interfaces, and Node.js with Express.js as robust solutions for RESTful API development. Collectively, these technologies support real-time operations and scalable system architectures suitable for event management platforms [5], [6].

Cloud-based systems are also extensively discussed in the literature, with emphasis on data security, automated backups, scalability, and multi-department accessibility. MongoDB Atlas is frequently cited for its distributed architecture, high availability, and suitability for academic information systems [5].

Another emerging trend in academic event management is QR-code-based attendance tracking. Research confirms that QR-enabled systems significantly reduce proxy attendance, accelerate verification processes, and enhance data accuracy. Such systems are increasingly adopted in both academic and corporate environments due to their reliability and efficiency [3], [6].

Despite the availability of multiple event management platforms, the literature consistently identifies several gaps, including the absence of AI-driven personalization, limited automation, lack of academic-focused design, weak integration between event creation and certificate distribution, insufficient analytics, and ineffective communication mechanisms [1], [3], [4].

EventEase aims to bridge these gaps by integrating AI-based features, automated workflows, modern web technologies, and cloud infrastructure to deliver a comprehensive, scalable, and future-ready academic event management solution.

METHODOLOGY

The methodology followed for developing EventEase is divided into a few clear and systematic steps. Each step ensures that the system is built in an organized and efficient manner.

1. Requirement Collection

The first step was to understand the problems in the existing event management process. Discussions with students, organizers, and faculty helped identify requirements such as online registration, attendance tracking, event listing, and certificate generation.

2. System Design

After collecting requirements, the system structure was designed. This included creating flowcharts, database schemas, and layout designs for the user interface. The design helped in planning how the system would work and how different modules would interact with each other.

3. Development

The development was carried out using the MERN stack:

- **Frontend (React.js)** for creating user interfaces
- **Backend (Node.js + Express.js)** for APIs and server logic
- **Database (MongoDB Atlas)** for storing event and user data

Each module like registration, event creation, login, attendance, and certificate generation was developed one by one.

4. Testing

After development, the system was tested to ensure that all features worked correctly. Testing helped find and fix errors, check input validation, and improve user experience.

5. Deployment

Once testing was completed, the system was deployed on a cloud platform so that users can access it online. The database was also hosted securely on MongoDB Atlas.

6. Maintenance

After deployment, feedback was taken from users to make improvements and plan future enhancements.

PROPOSED SYSTEM

1. Introduction to the Proposed System

The proposed system, EventEase, is a fully digital, AI-driven event management platform designed to overcome the limitations of manual event handling. It centralizes all event-related activities—creation, registration, attendance, certificate generation, and reporting—into one automated and user-friendly interface accessible to students, organizers, and administrators.

2. Centralized Event Management Platform

The system provides a unified portal where all events are listed with details such as date, time, venue, description, and registration deadlines. This ensures that students can easily access information and stay updated on all activities within the institution.

3. Role-Based Secure Authentication

EventEase supports three roles: **Student, Organizer, and Administrator**. Each login type has its own dashboard and permissions.

- Students view and register for events.
 - Organizers manage event creation, attendance, and certificates.
 - Admins approve events and monitor overall system operations.
- Security is ensured through JWT-based authentication and encrypted data storage.

4. Online Registration System

The proposed system eliminates paper-based registration by offering instant digital registration. Students can register for events with one click, and organizers can view participant lists in real time. This reduces errors and speeds up the process.

5. QR-Based Attendance Automation

Attendance is handled using QR codes generated for each registered participant. Organizers scan these codes during the event, and attendance is logged instantly into the database. This ensures accuracy, eliminates proxy entries, and saves time for large events.

6. AI-Powered Certificate Generation

EventEase integrates an AI-based certificate automation module. After the event, the system automatically generates personalized certificates with participant names, event details, and dates. Students can download them immediately from their dashboard. This removes human error and reduces manual workload.

7. Real-Time Dashboards and Analytics

The system provides interactive dashboards with real-time statistics:

- Total registrations
- Attendance percentage
- Event reach and popularity
- Department-wise participation

These insights help administrators make informed decisions and improve event planning in the future.

8. Improved Communication Through Notifications

EventEase includes automated notifications to inform students about event deadlines, updates, schedule changes, and certificate availability. This reduces confusion and improves engagement.

FLOW DIAGRAM

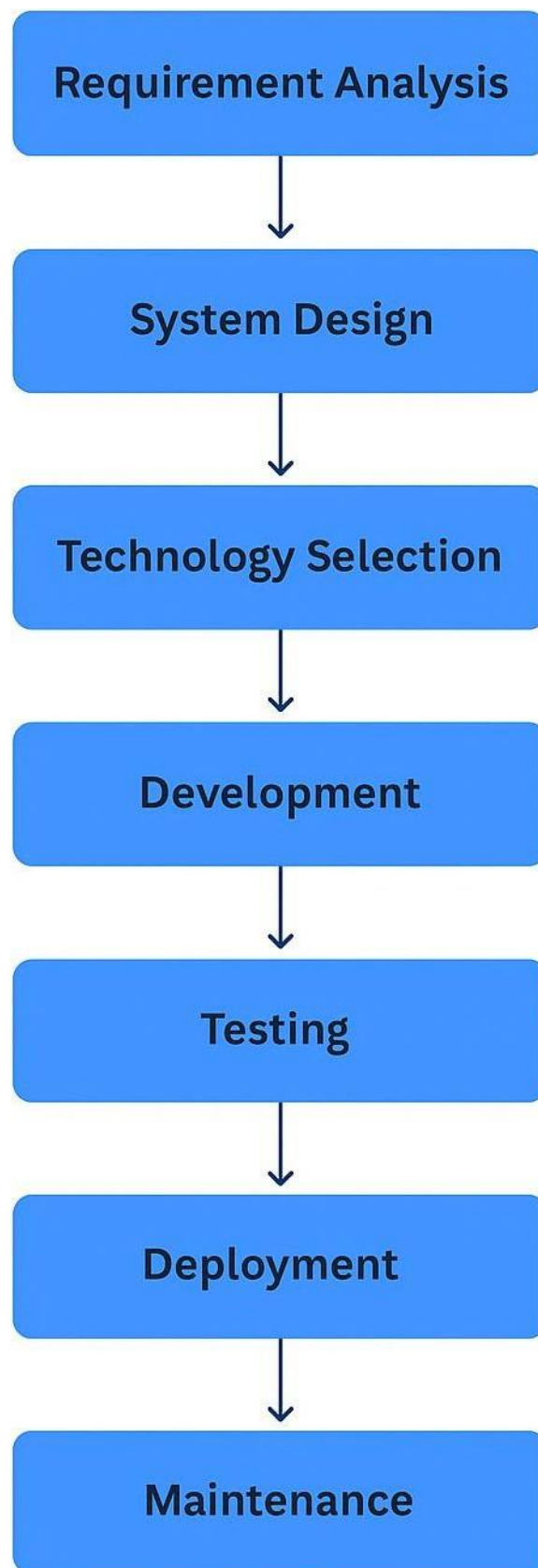


Figure: Methodology Used for EventEase Development

RESULT

1. System Functionality Outcomes

The EventEase system successfully implemented all major functional modules, including event creation, user authentication, online registration, QR-based attendance, and certificate generation. Testing showed that each module worked smoothly and produced accurate results. Students were able to register instantly, organizers could manage events without difficulty, and administrators could approve events efficiently. All core features such as event listing, notifications, and role-based dashboards operated as expected, demonstrating the reliability and completeness of the system's functionality.

2. Performance and Efficiency Results

Performance testing indicated that the system responded quickly even under multiple simultaneous requests. Database operations such as fetching events, updating attendance, and generating certificates executed in minimal time due to optimized MERN-based architecture. The shift from manual operations to automated workflows significantly increased efficiency—registration time dropped from minutes to seconds, and certificate generation became almost instantaneous. The overall performance showed that the system is capable of supporting large-scale student participation and multiple concurrent events.

3. User Experience and Interface Evaluation

Users reported that the interface was simple, clean, and easy to navigate. The React-based frontend provided smooth transitions, responsive design, and consistent layouts across devices. Students could quickly find relevant events, and organizers found dashboards intuitive for managing data. The logical placement of features ensured minimal learning effort. Feedback indicated high user satisfaction due to the system's simplicity, clarity, and modern UI.

4. Automation and Accuracy Findings

Automation features performed exceptionally well during testing. The QR attendance module accurately recorded entries with zero duplication or proxy concerns. The AI-powered certificate generator produced error-free certificates with correct formatting and participant details. Automated notifications ensured timely communication and reduced misunderstandings. Overall accuracy improved drastically compared to manual systems, reducing human error and administrative workload.

5. Overall System Effectiveness

The combined results indicate that EventEase is highly effective in solving the challenges of traditional event management. It reduces manual tasks, accelerates event workflows, enhances communication, and increases transparency. The system is stable, scalable, user-friendly, and reliable. It successfully supports the complete event lifecycle—from creation to post-event certification—making it a practical and impactful solution for colleges.

CONCLUSION

1. Summary of the Developed System

It integrates event creation, registration, attendance, communication, and certificate generation into one unified platform. The system ensures smooth coordination among students, organizers, and administrators through role-based access and easy-to-use dashboards.

2. Achievement of Project Goals

All project objectives were successfully achieved. The system eliminates manual errors, reduces paperwork, and provides automated features such as QR attendance and AI-based certificate creation. It also improves transparency and user engagement through real-time updates and notifications. The technology stack used enables fast performance and reliable data handling.

3. Benefits and Improvements Over Existing Methods

Compared to traditional event management, EventEase drastically improves efficiency, speed, and accuracy. Registrations take seconds, certificates generate instantly, and attendance is captured accurately with QR codes. Unlike Google Forms or manual sheets, EventEase provides centralized data storage, dashboards, analytics, automated processes, and seamless communication—all in one platform.

4. Limitations of the Current Implementation

Although the system is functional and reliable, certain limitations still exist. The system currently lacks features such as integrated payment gateways for paid events, cross-college collaboration, advanced AI predictions, and a mobile application. Additionally, it depends on internet connectivity, which may affect usage in limited-network environments.

5. Future Enhancements and Final Remarks

Future upgrades may include a mobile app, integrated payment processing, chatbot support for event queries, AI-based turnout prediction, and feedback analysis for event improvement. In conclusion, EventEase successfully modernizes college event management and demonstrates how automation and AI can transform administrative processes. It sets a strong foundation for smarter, scalable, and future-ready event ecosystems.

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